

SchuF Blowdown Valves



SchuF 

Blow-down Valves Continuous, Angle-Type

ASME 150-1500

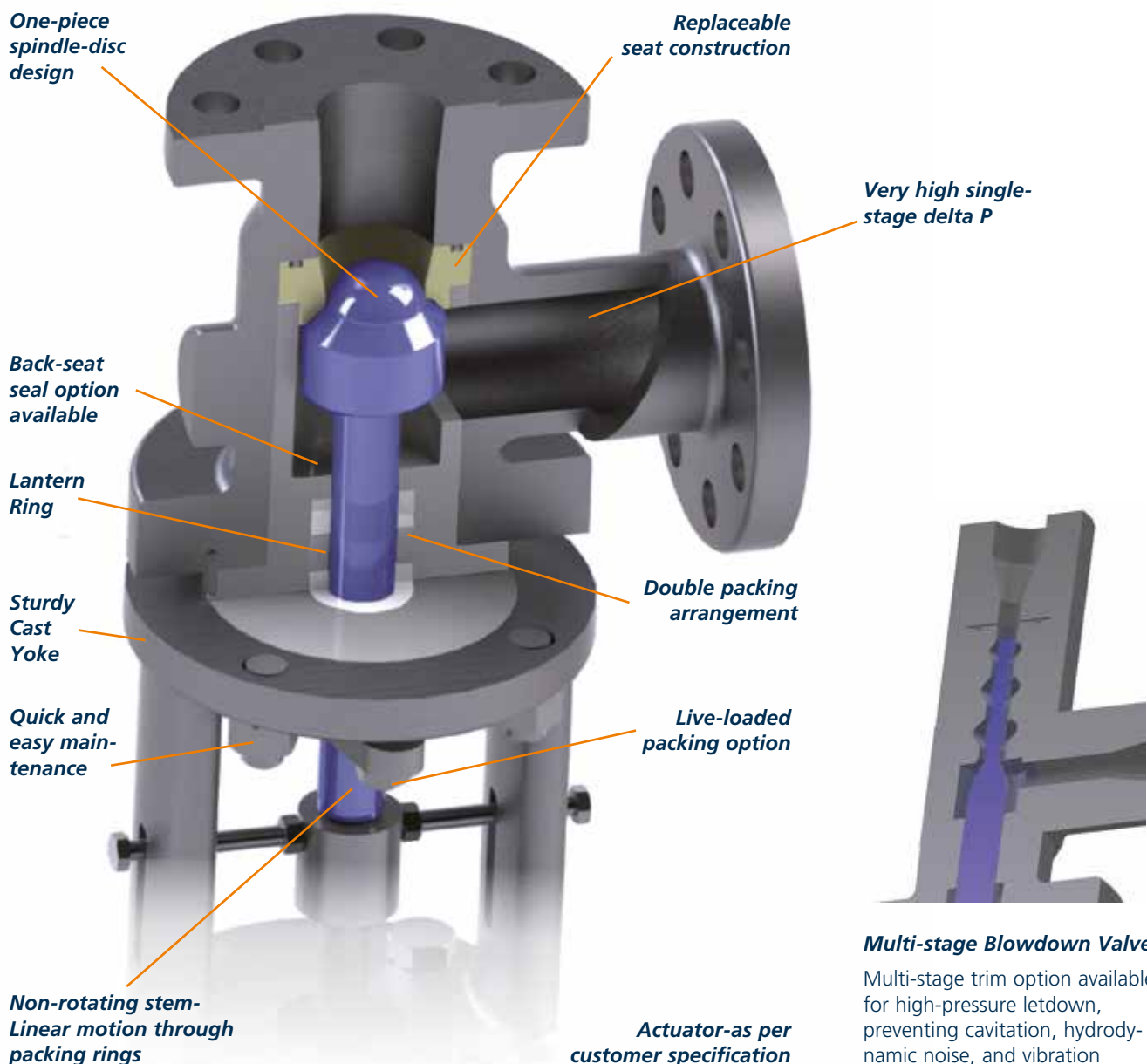
Type 71

The primary purpose of blow down valves is the removal of dirt, scale and sediment from boilers or associated pipelines in order to maintain boiler integrity and efficiency. Steam boiler systems in principal utilise a continuous blow-down valve to control the removal of dissolved elements (salt and other chemicals) at the top of the boiler water, and also a discontinuous blow-down valve to remove solids and accumulated sediments from the pipelines.

Code requires that residue cannot accumulate in these valves, and SchuF manufactures continuous blow-down valves that minimise such accumulations and remove dissolved solids and particulate in a continuous process so that the minimum quantity of water is discharged from the boiler.

Specific advantages include:

- Low fugitive emissions can be achieved with the double packing arrangement.



Blow-down Valves Continuous, Angle-Type

ASME 150-1500

Type 71

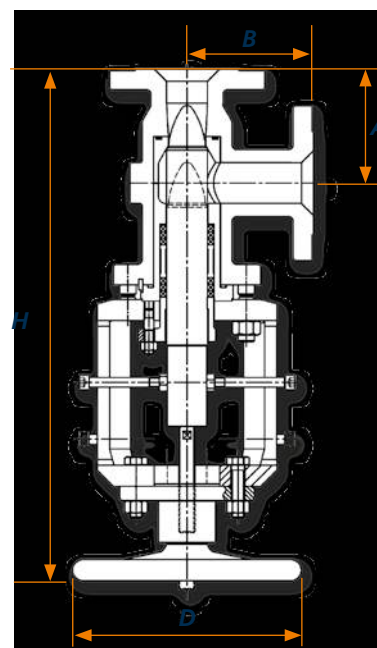
Dimensions mm/Size

Rating →	150#				300#			
Size ↓	A	B	H	D	A	B	H	D
1" DN025	100	110	435	Ø160	103	113	435	Ø160
1½" DN040	118	128	500	Ø160	122	132	500	Ø160
2" DN050	125	150	565	Ø225	128	153	565	Ø160
3" DN080	170	225	800	Ø280	174	229	800	Ø225
4" DN100	205	285	915	Ø315	213	293	910	Ø280
6" DN150	296	416	1210	Ø360	307	427	1210	Ø315

Additional Dimensions for larger sizes and higher pressure ratings available on request

CV Values/Size

Size	Cv (300#)
1" DN025	18
1½" DN032	42
2" DN050	65
2½" DN065	102
3" DN080	160
4" DN100	270
6" DN150	565



Description:	Standard:	Optional:
Size:	DN 25 (1") to DN 600 (24")	Larger on request
Pressure Class:	ASME 150 to ASME 600	Higher on request
Body Material:	A216 WCB, A217 WC6, A351	316 SS, Super-Duplex, Incoloy, Hastelloy, others on Request
Trim:	Stainless Steel, A182 F6	As above, plus fully stellited
Sealing:	Metal-to-metal	Stelliting, other Hard-facing
Seal to Atmosphere:	Graphite Packing	TA-Luft, Low Emissions
Connections:	Flanges, Butt-weld	Others on Request
Standards / Cert.:	ASME B16.34, API 598 & 623, EN 12266, Dir. PED 97/23/EC	NACE MR0175, others on request
Actuation:	Manual	Bevel-Gear, Electric, Pneumatic

Blow-down Valves Continuous, Y-Type

ASME 150-1500

Type 50SR

The primary purpose of blow down valves is the removal of dirt, scale and sediment from boilers or associated pipelines in order to maintain boiler integrity and efficiency. Steam boiler systems in principal utilise a continuous blow-down valve to control the removal of dissolved elements (salt and other chemicals) at the top of the boiler water, and also a discontinuous blow-down valve to remove solids and accumulated sediments from the pipelines.

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Specific advantages include:

- Low fugitive emissions can be achieved with the double packing arrangement.

Quick and easy maintenance

Non-rotating stem-Linear motion through packing rings

Live-loaded packing option

Double packing arrangement

One-piece spindle-disc design

Replaceable seat construction

Actuator-as per customer specification

Sturdy Cast Yoke

Lantern Ring

Back-seat seal option available

Very high single-stage delta P

Blow-down Valves Continuous, Y-Type

ASME 150-1500

Type 50SR

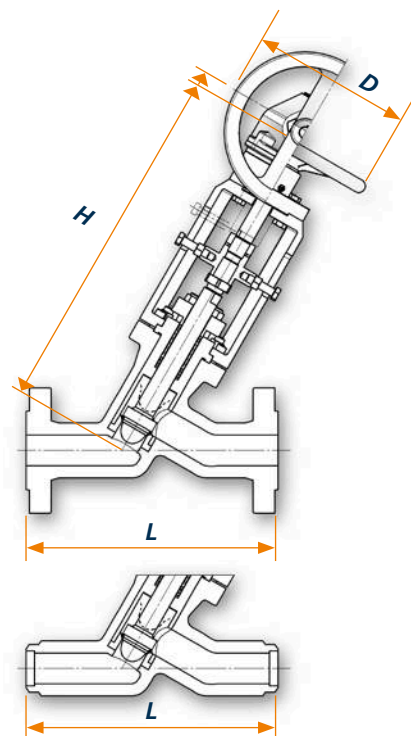
Dimensions mm/Size

Size ↓	A	B	H	D	A	B	H	D
1" DN025	190	130	360	Ø160	196	133	360	Ø160
1½" DN040	286	198	440	Ø160	294	202	440	Ø160
2" DN050	350	245	530	Ø225	356	248	530	Ø160
3" DN080	530	370	745	Ø280	538	374	745	Ø225
4" DN100	630	445	845	Ø360	646	453	840	Ø280
6" DN150	902	636	1110	Ø360	924	647	1110	Ø315

Additional Dimensions for larger sizes and higher pressure ratings available on request

CV Values/Size

Size	Cv (600#)
1" DN025	25
2" DN050	70
2½" DN065	95
3" DN080	110
4" DN100	190
6" DN150	390



Description:	Standard:	Optional:
Size:	DN 25 (1") to DN 600 (24")	Larger on request
Pressure Class:	ASME 150 to ASME 600	Higher on request
Body Material:	A216 WCB, A217 WC6, A351	316 SS, Super-Duplex, Incoloy, Hastelloy, others on Request
Trim:	Stainless Steel, A182 F6	As above, plus fully stellite
Sealing:	Metal-to-metal	Stellite, other Hard-facing
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Connections:	Flanges, Butt-weld	Others on Request
Standards / Cert.:	ASME B16.34, API 598 & 623, EN 12266, Dir. PED 97/23/EC	NACE MR0175, others on request
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Blow-down Valves Discontinuous/Intermittent, Angle Type

ASME 150-1500

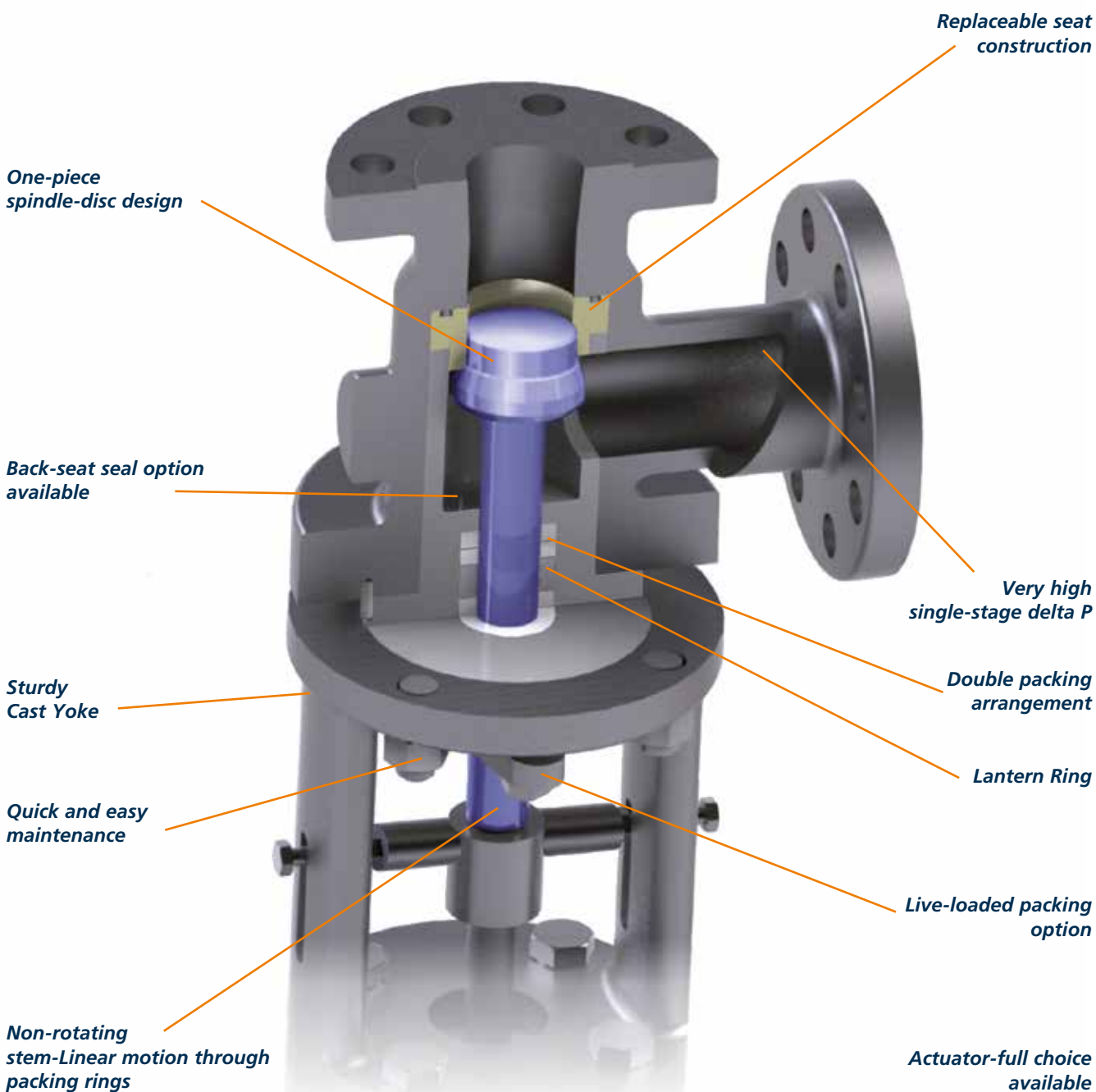
Type 71

The primary purpose of discontinuous/intermittent blow-down valves is the removal of dirt, scale and sediment from boilers in order to maintain boiler integrity and efficiency. Code requires that residue cannot accumulate in these valves.

SchuF manufactures discontinuous blow-down valves that minimise such internal accumulations so that the minimum quantity of water is discharged from the boiler.

Specific advantages include:

- Low fugitive emissions can be achieved with the double packing arrangement.



Blow-down Valves Discontinuous/Intermittent, Angle Type

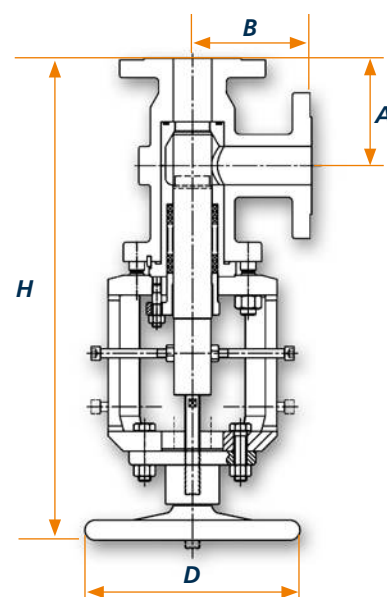
ASME 150-1500

Type 71

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3" DN080	170	225	825	Ø315	174	229	825	Ø315
4" DN100	205	285	920	Ø315	213	293	920	Ø315
6" DN150	296	416	1225	Ø400	307	427	1225	Ø400

Additional Dimensions for larger sizes and higher pressure ratings available on request



CV Values/Size

Size	Cv (300#)
1" DN025	19
1½" DN032	44
2" DN050	67
2½" DN065	102
3" DN080	161
4" DN100	268
6" DN150	563

Description:	Standard:	Optional:
Size:	DN 25 (1") to DN 600 (24")	Larger on request
Pressure Class:	ASME 150 to ASME 600	Higher on request
Body Material:	A216 WCB, A217 WC6, A351	316 SS, Super-Duplex, Incoloy, Hastelloy, others on Request
Trim:	Stainless Steel, A182 F6	As above, plus fully stellited
Sealing:	Metal-to-metal	Stelliting, other Hard-facing
Seal to Atmosphere:	Graphite Packing	TA-Luft, Low Emissions
Connections:	Flanges, Butt-weld	Others on Request
Standards / Cert.:	ASME B16.34, API 598 & 623, EN 12266, Dir. PED 97/23/EC	NACE MR0175, others on request
Actuation:	Manual	Bevel-Gear, Electric, Pneumatic

Blow-down Valves Discontinuous/Intermittent, Y-Type

ASME 150-1500

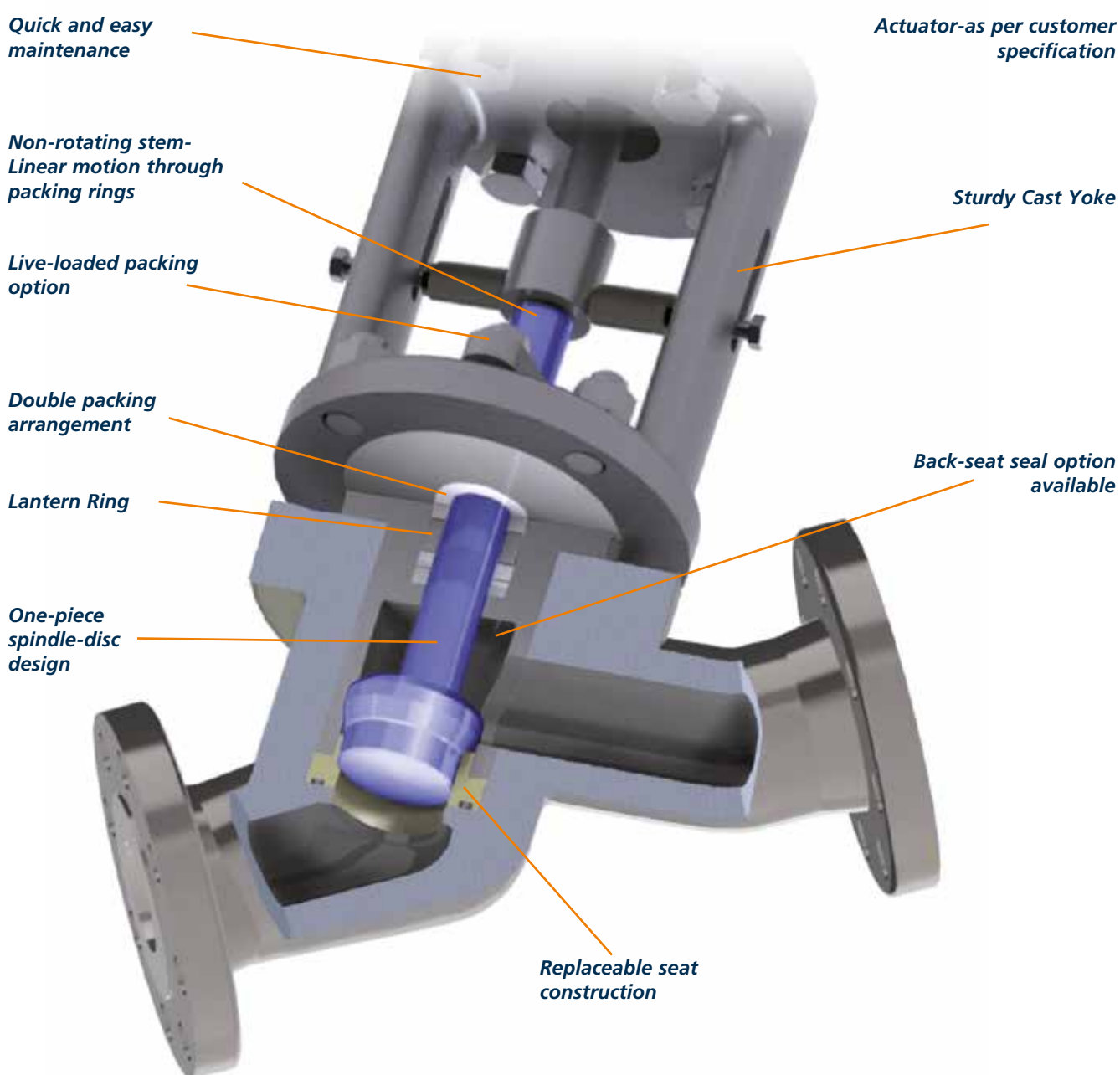
Type 50ST

The primary purpose of blow down valves is the removal of dirt, scale and sediment from boilers in order to maintain boiler integrity and efficiency. Code requires that residue cannot accumulate in these valves

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Blow-down Valves Discontinuous/Intermittent, Y-Type

ASME 150-1500

Type 50ST

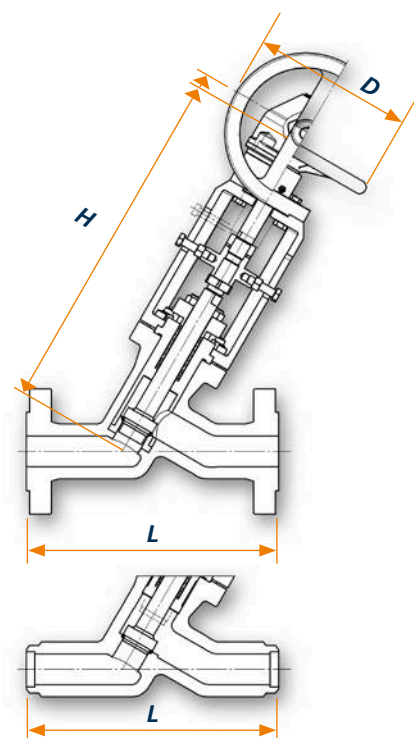
Dimensions mm/Size

Size ↓	A	B	H	D	A	B	H	D
1" DN025	190	130	395	Ø160	196	133	395	Ø160
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4" DN100	190
6" DN150	390



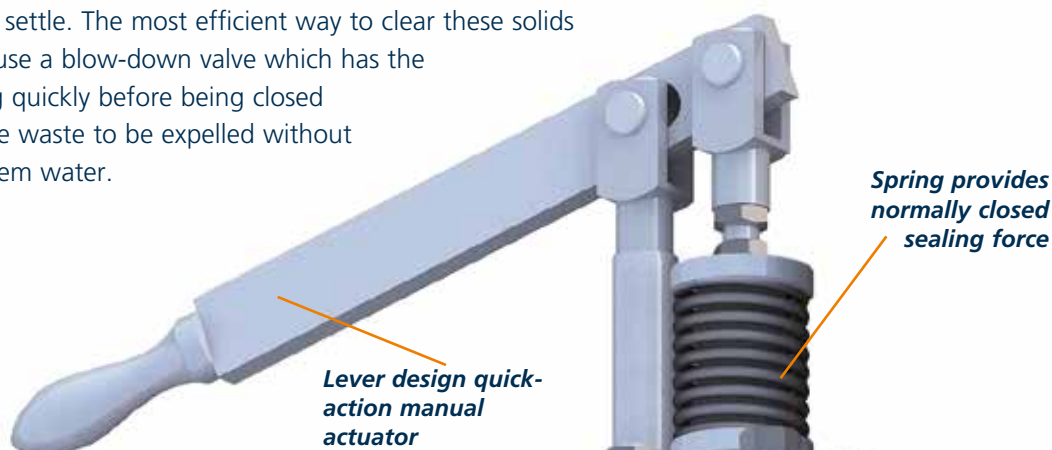
Description:	Standard:	Optional:
Size:	DN 25 (1") to DN 150 (6")	Larger on request
Pressure Class:	ASME 150 to ASME 1500	Higher on request
Body Material:	A216 WCB, A217 WC6, A351	316 SS, Super-Duplex, Incoloy, Hastelloy, others on Request
Trim:	Stainless Steel, A182 F6	As above, plus fully stellited
Sealing:	Metal-to-metal	Stelliting, other Hard-facing
Seal to Atmosphere:	Graphite Packing	TA-Luft, Low Emissions
Connections:	Flanges, Butt-weld	Others on Request
Standards / Cert.:	ASME B16.34, API 598 & 623, EN 12266, Dir. PED 97/23/EC	NACE MR0175, others on request
Actuation:	Manual	Bevel-Gear, Electric, Pneumatic

Blow-down Valves Quick-Action Type

Discontinuous Valve

Quick-Action Discontinuous Blow-Down Valve

- Steam boiler systems in principal utilise a discontinuous blow-down valve to remove solids and accumulated sediments from the lower levels of boiler vessels. This valve type is placed at the low point of the system where the solids will settle. The most efficient way to clear these solids and sediments is to use a blow-down valve which has the capability of opening quickly before being closed again. This allows the waste to be expelled without excessive loss of system water.



- This Discontinuous Blow-Down valve (right) utilises a Dead-Man's-Handle actuator, which can be opened and closed virtually instantaneously. The valve's sealing force is supplied by the actuator spring. The simple lever action allows an operator to perform the blow-down operation quickly and efficiently



Disc rising design

Hard-faced seating option

Access port

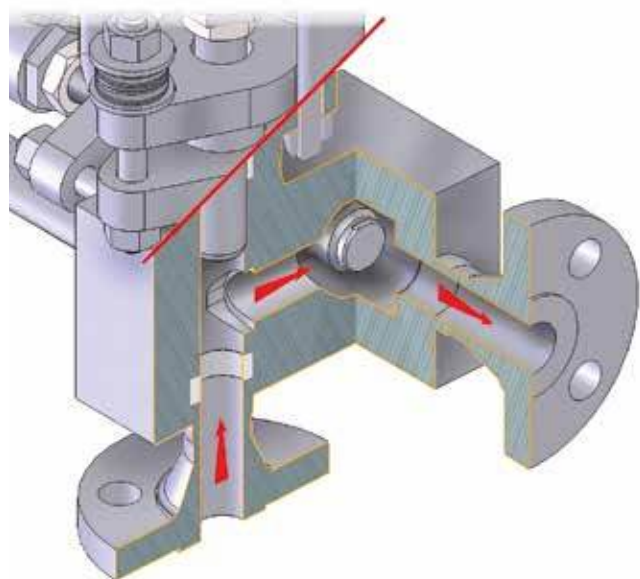
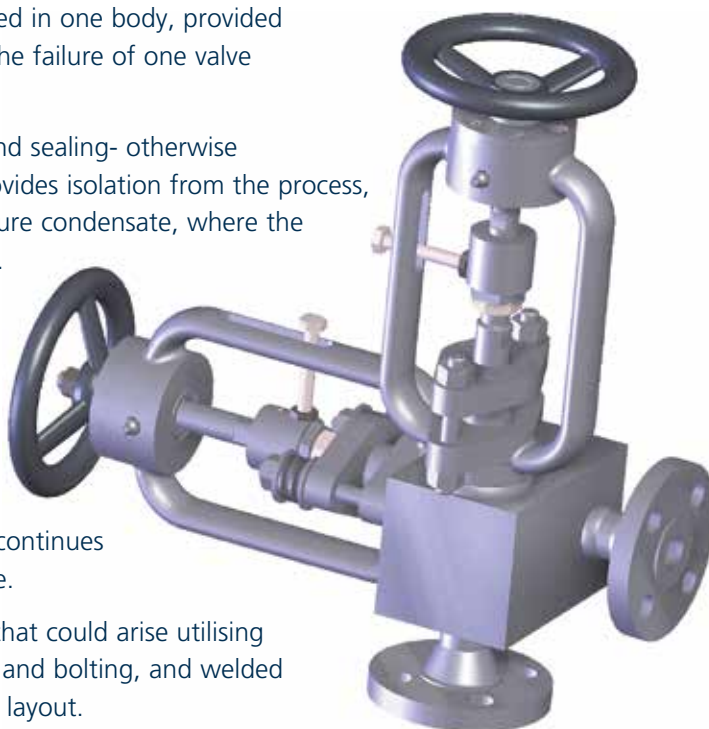
- These Blow-Down valves can also be supplied with powered actuation, such as this pneumatically operated version.

Blow-down Valves Combination Type

Tandem Blow-Down Valve

Tandem Blow-Down Valve

- ASME Code for Pressure Piping, B31.1, instructs that blow-down piping systems from boiler water spaces require two shutoff valves in series. In addition, one or both of these valves must be slow-opening (requiring at least five 360° turns of the operating mechanism to change from being fully-closed to fully-opened).
- Code states that these two valves may be combined in one body, provided that the two valves operate individually and that the failure of one valve does not affect the operation of the other.
- This valve combines the two functions- blowing and sealing- otherwise carried out by two individual valves. One valve provides isolation from the process, and the other valve is used to blow off high-pressure condensate, where the medium could contain dissolved solids and sludge.
- The isolation valve will not be exposed to the same corrosion as the blow-down valve, as no medium will flow through the system while the blow-down valve is still closed (step 2 below). The blow-down valve will experience strong abrasive flow immediately upon opening. Its performance can be monitored by how well it continues to seal following the opening of the isolation valve.
- This Tandem Valve construction eliminates issues that could arise utilising separate valves with flanged connections, gaskets and bolting, and welded piping connections- and does so in a space-saving layout.



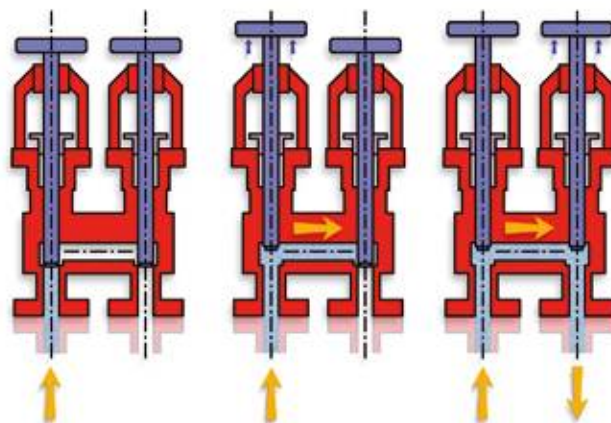
3D Cutaway View

Tandem Blow-Down Valve Operation/Schematic

1. System Closed

2. Isolation valve opened first; Blow-down valve now provides sealing

3. Blow-down valve opened, system draining begins



Note- Closure is the reverse of the steps outlined above, i.e. Blow-Down valve closed first.

Customer Enquiry Sheet

Part 1 – Your Company Information

Name:	Title:
Company:	Telephone:
E-mail:	Fax:

Part 2 – Your Order Information

General:

Quantity:	Application:
Tag No.:	Project Name:
Valve Model:	Project No.:

Part 3 – Valve and Process Information

Valve Information:

Valve Inlet Size (DN/Inch):	Valve Pressure Rating:
Valve Outlet Size (DN/Inch):	Valve Delta P:
Inlet Flanged/Weld-End:	Outlet Flanged/Weld-End:
Operating Temp.:	Operating Pressure:
Design Temp.:	Design Pressure:
Body Material:	Trim Material:
Medium:	Fugitive Emissions/Clean Air:
Certificates Etc.:	Firesafe Rating:
Actuator Type:	Air Supply Pressure:
Air-Fail Position:	Manual Override:

Part 4 – Further Notes/Topics/Info

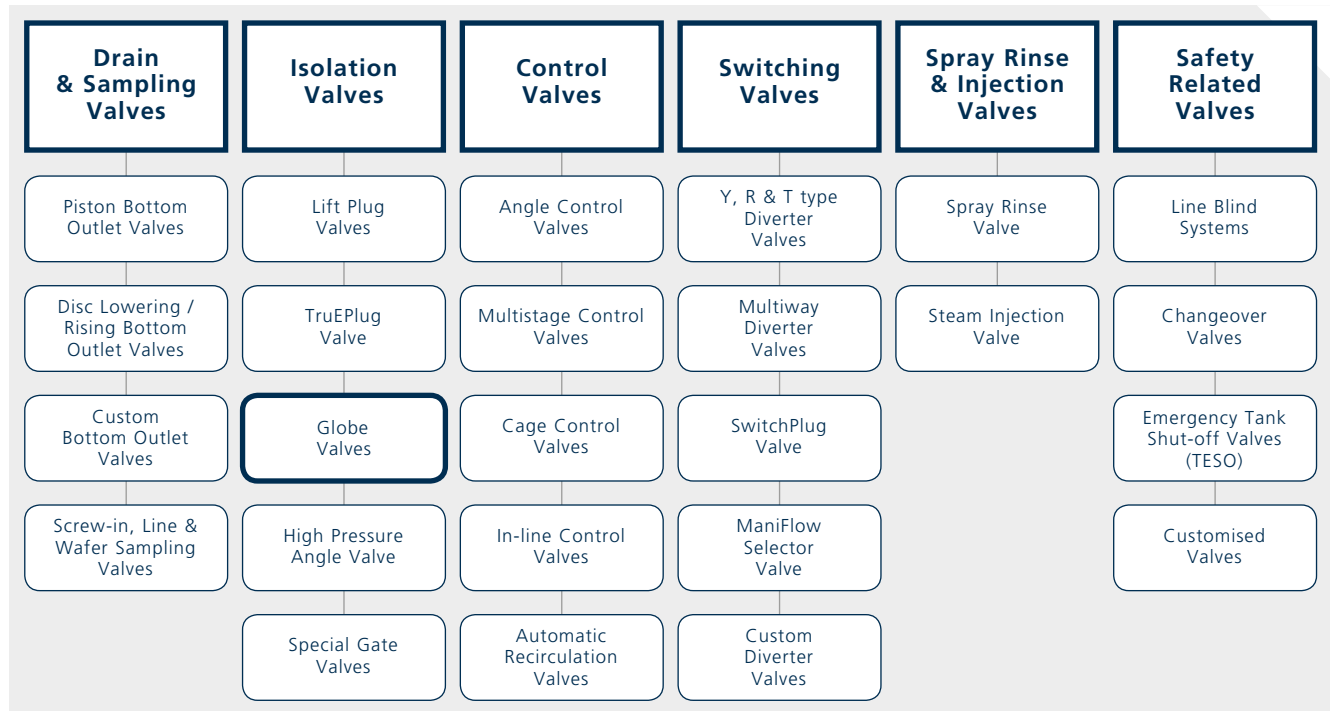
Product Portfolio Overview

SchuF Fetterolf has delivered over one million valves during its 100 year history to a wide variety of industries in over 50 countries worldwide.

Headquartered near Frankfurt in Germany, the company has additional design and manufacturing centres in Italy, India, Ireland, UK and the USA.

The SchuF group has sales and agent offices covering almost every country in the world.

We manufacture valve products that control, isolate, divert, and sample liquids, gases, powders, and slurries. Our product range of engineered, customised valves includes:

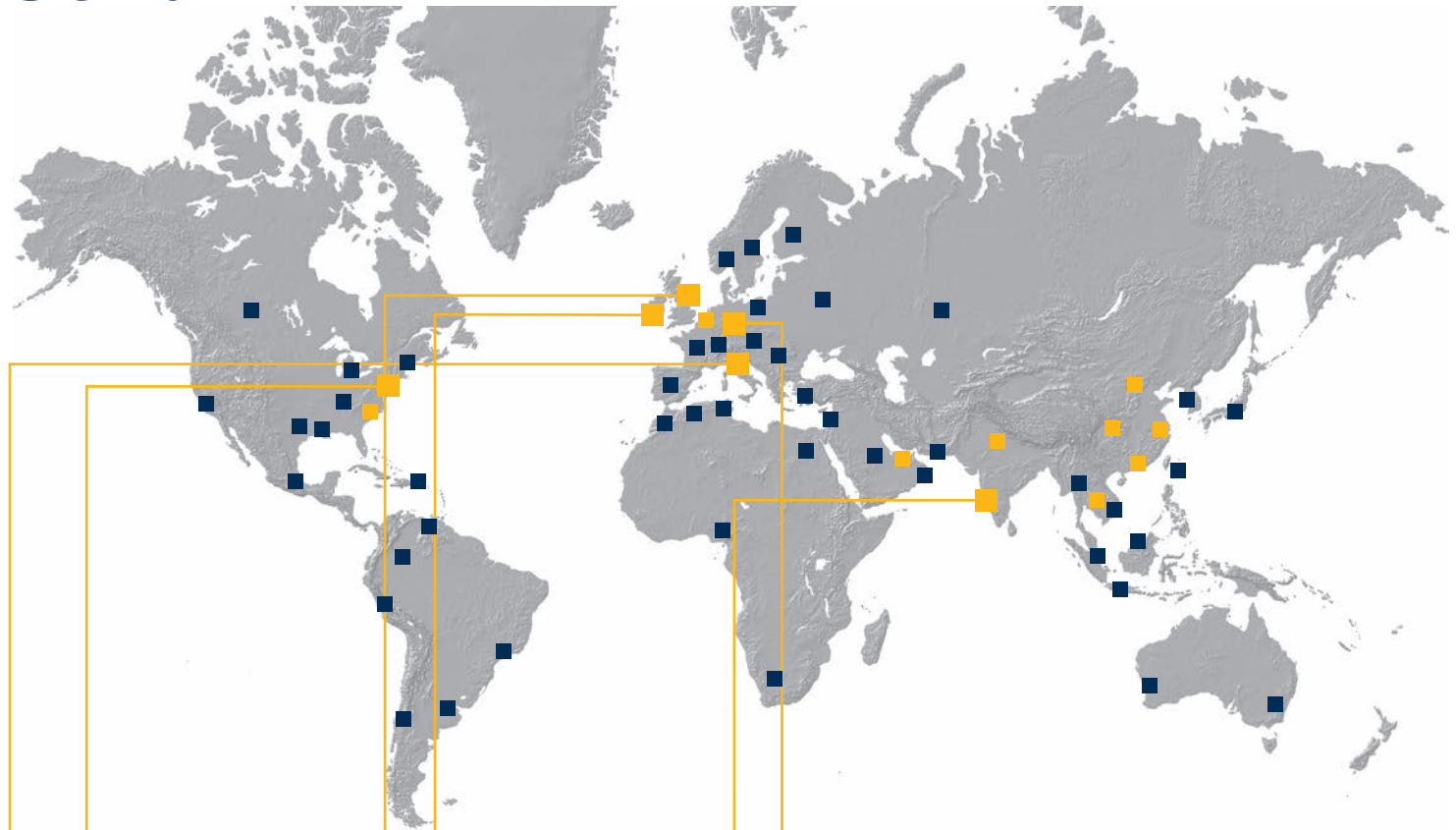


Globe Valve Client List:

- | | |
|----------------------------|-----------------------------|
| ■ AirLiquide | ■ Indorama Synthetics |
| ■ Aluminium Oxid Stade | ■ Invista |
| ■ BASF | ■ Jindal Steel & Power |
| ■ BAYER | ■ Lanxess |
| ■ BP | ■ Li Peng Enterprise |
| ■ Chang Chun Petrochemical | ■ Monsanto |
| ■ Cheil | ■ Reliance |
| ■ Clariant | ■ SABIC |
| ■ DSM Hydrocarbons | ■ Samsung |
| ■ Du Pont | ■ Saudi Aramco |
| ■ Dow Chemical | ■ Shinkong Synthetic Fibers |
| ■ Eastman Chemical | ■ Tianjin Petrochemical |
| ■ Evonik | ■ Uhde Inventa-Fischer |
| ■ Far Eastern Industries | ■ Vinnolit |
| ■ Foster Wheeler | ■ Zhejiang Hengyi Polymer |



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